

EMCH 308 Semester Project

Design an I-beam with the dimensions and access holes shown in Fig. 1 to meet the following specifications:

1. Total weight less than or equal to 1 lbf
2. Safety factor greater than or equal to 1.25 for a 5,000 lbf load applied in 3-pt bending
3. Total manufacturing cost is less than or equal to \$600

The goal is to meet the performance requirements (items 1 and 2) while minimizing the manufacturing costs. You must use the project costing spreadsheet to estimate costs and select material stock from the three options included on the spreadsheet.

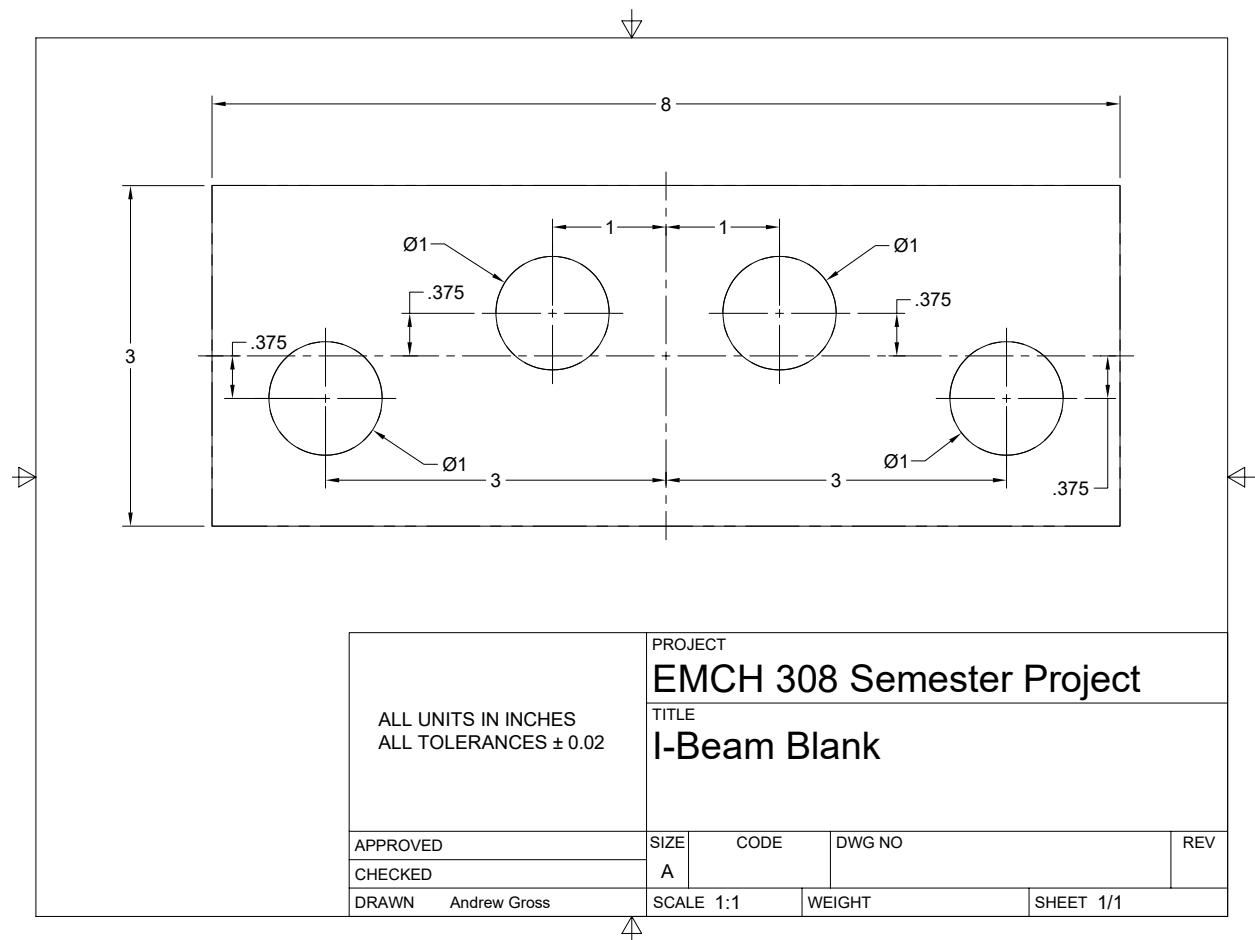


Figure 1: Profile of the I-beam indicating the 8" span, 3" height, and size and location of four through holes. Removal of additional material is allowed.

Details:

1. Load will be applied with a centrally located steel roller that has a 0.984" diameter.
2. Beams will be supported by two steel rollers, each with 0.394" diameter, that have a center to center distance of 8 inches.
3. Satisfaction of the safety factor requirement will be measured by loading the structure to 6,250 lbf and checking for intersection of the load vs. displacement curve with a line offset from the linear-elastic behavior by a displacement of 0.02 inches.